Lecture 7. Combinatorics

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Review: Equally likely outcomes

- When the sample space Ω consists of n possible outcomes, $\omega_1, \ldots, \omega_n$, each having the same probability.

$$\sum_{k=1}^{n} P\{\omega_k\} = P\{\Omega\} = 1$$

then $P\{\omega_k\} = \frac{1}{n}$ for each k.

- Further, the probability of any event E consisting of t outcomes, equals

$$\mathrm{P}\{\mathrm{E}\} = \sum_{\omega_k \in E} \left(\frac{1}{n}\right) = \frac{t}{n} = \frac{number\ of\ outcomes\ in\ E}{number\ of\ outcomes\ in\ \Omega} = \frac{N_E}{N_T}$$

Example

A card is drawn from a bridge 52-card deck at random. Compute the probability that the selected card is a spade.

Solution 1: The sample space consists of 52 equally likely outcomes cards. Among them, there are 13 favorable outcomes.

$$P\{spade\} = \frac{13}{52} = \frac{1}{4}$$

Solution 2: The sample space consists of 4 equally likely outcomes-suits: clubs, diamonds, hearts and spades. Among them, one outcome is favorable-spades.

$$P\{spade\} = \frac{1}{4}$$

Example

A young family plans to have two children. What is the probability of two girls?

Wrong solution: There are 3 possible families with 2 children: two girls, two boys and one of each gender. Therefore, the probability of two girls is $\frac{1}{3}$.

Correct solution: Each child is equally likely to be a boy or a girl. Genders of the two children are independent. Therefore

$$P\{two\ girls\} = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{4}$$

Permutation

- A permutation is a selecting k items from a set of n items where order matters and repeating is not allowed.

Permutation without replacement:

$$P(n,k) = \frac{n!}{(n-k)!}$$

where $n! = n \cdot (n-1) \cdot (n-2) \cdot \ldots \cdot 2 \cdot 1$.

Permutation with replacement

$$P_r(n,k) = n^k$$

Combination

- Selecting k items from a set of n items where order does not matter and repeating is not allowed is called a combination.

Combination without replacement:

$$C(n,k) = \binom{n}{k} = \frac{n!}{k!\,(n-k)!}$$

Combination with replacement:

$$C_r(n,k) = \binom{n+k-1}{k} = \frac{(n+k-1)!}{k!\,(n-1)!}$$

Example

One can create an alphabet with 10 digits numbers, 26 lowercase, and 26 capital letters. How many 8-character password can be created?

With replacement case:

$P_r(62,8) = 62^8 = 62 * 62 * 62 * 62 * 62 * 62 * 62 * 62 = 2.18 * 10^{14}$

Without replacement case:

$$P(62,8) = \frac{62!}{(62-8)!} = 1.36 * 10^{14}$$

Example

How many ways can ten students be seated in a classroom with 15 chairs?

Students are distinguishable, and each student needs a separate seat. The number of possible allocations is the number of permutations without replacement.

$$P(15,10) = \frac{15!}{(15-10)!} = 15 * 14 * 13 * 12 * 11 * 10 * 9 * 8 * 7 * 6 = 1.09 * 10^{10}$$

Example

An antivirus software reports that three folders out of 10 are infected. How many possibilities are there?

(A, B, C), (B, A, C) and (C, B, A) represent the same outcome. The order is not important. The software clearly detected three different folders. Thus, it is sampling without replacement.

$$C(10,3) = \binom{10}{3} = \frac{10!}{3!\,(10-3)!} = 120$$

Example (Your Turn)

a) How many ways to choose 7 people to form a committee (all members play the same role) from a class of 50 people?

$$C(50,7) = \binom{50}{7} = \frac{50!}{7!(50-7)!} = 9.99 * 10^7$$

b) How many ways to choose 3 people to form a cabinet (president, vice president, treasurer) from a class of 50 people?

$$P(50,3) = \frac{50!}{(50-3)!} = 1.18 * 10^5$$

Example (Your Turn)

c) How many car license plates are available if the rule is three letters followed by four numbers.

Total number = $26*26*26*10*10*10*10=1.75*10^8$

Fundamental Theorem of Counting （FTC)

FTC： Suppose you want to calculate the number of different ways that "task" can be performed.

1. The task can be broken down into k steps, and

2. Step i can be done in n_i different ways regardless of how the previous steps have been performed, and

3. Each different way of performing the steps gives a different way of performing the task, then the task can be done in $n_1 * n_2 *...,* n_k$ different ways.

Example

There are 20 computers in a store. Among them, 15 are brand new and 5 are refurbished. Six computers are purchased for a student lab. From the first look, they are indistinguishable, so the six computers are selected at random. Compute the probability that among the chosen computers, two are refurbished.

Let A:={there are two refurbished computers among six selected computers}

$\#(\Omega) = \binom{20}{6} = 38760$

Then we use FTC to calculate #(A)

Step 1: Choose two from five refurbished computers $\binom{5}{2}$

Step 2: Choose four from fifteen brand new computers $\binom{15}{4}$

$\#(A) = \binom{5}{2}\binom{15}{4} = 10*1365 = 13650$

$$P\{two\ refurbished\ computers\} = \frac{\#(A)}{\#(\Omega)} = \frac{\binom{5}{2}\binom{15}{4}}{\binom{20}{6}} = \frac{13650}{38760} = 0.352$$

Example

Draw 5 cards from a well-shuffled deck of 52. What is the probability of full house? A "full house means (three of a kind + a pair).

Let A:={draw a full house}

#(Ω) = $\binom{52}{5}$=2,598,960

Then we use FTC to calculate #(A)

Step 1: Choose value to be repeated 3 times $\binom{13}{1}$

Step 2: Choose 3 cards of that value $\binom{4}{3}$

Step 3: Choose value for pair $\binom{12}{1}$

Step 4: Choose 2 cards of that value $\binom{4}{2}$

#(A) = 13*4*12*6=3744

$$P(A) = \frac{\#(A)}{\#(\Omega)} = \frac{3744}{2598960} = 0.0014$$

Example (Your Turn)

Draw 4 cards from a well-shuffled deck of 52. What is the probability of two pairs?

Example (Your Turn)

Draw 4 cards from a well-shuffled deck of 52. What is the probability of two pairs?

Let B:={draw two pairs}

$\#(\Omega) = \binom{52}{4} = 270{,}725$

Then we use FTC to calculate #(B)

Step 1: Choose value to be repeated 2 times $\binom{13}{1}$

Step 2: Choose 2 cards of that value $\binom{4}{2}$

Step 3: Choose another value to be repeated 2 times $\binom{12}{1}$

Step 4: Choose 2 cards of that value $\binom{4}{2}$

$$P(B) = \frac{\#(B)}{\#(\Omega)} = \frac{5616}{270725} = 0.0207$$